

# 衍生工具定价复习讲义

Derivatives Pricing Review Notes

中文主讲 + English key terms + assignment based practice

|                 |                                                           |
|-----------------|-----------------------------------------------------------|
| Course material | Lecture slides, syllabus, assignments, official solutions |
| Style           | Beginner friendly, formula focused, exam oriented         |
| Build           | XeLaTeX / ctexbook                                        |

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# How To Use This Review / 怎么用这份讲义

这份讲义不是把课件逐页翻译，而是把所有 lecture、assignment 和 solution 读完后，按考试复习逻辑重新组织。每一章都遵循同一个顺序：

1. **先讲直觉**：先回答“这个工具解决什么问题”，避免一上来就背公式。
2. **再讲公式**：每个公式都说明变量含义、适用条件和容易错的地方。
3. **再做题**：题目主要参考 assignments，assignment 1-3 用官方 solution 核对，assignment 4 根据 Lecture 12 的 binomial tree 方法推导。
4. **最后记英文表达**：考试题是英文时，最重要的是认出关键词，例如 forward price, hedge ratio, zero rate, swap rate, put-call parity, risk-neutral probability。

## 学习顺序建议

如果你是小白，先按这个顺序复习：Futures mechanics → Hedging → Interest rates → Forward/Futures pricing → Interest rate futures → Swaps → Options → Binomial trees。不要先背 option strategy 图形，否则容易只记形状不理解现金流。

## Source Coverage / 材料覆盖清单

| Material                               | Scale     | Used for                                                                                                    |
|----------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------|
| Derivatives Pricing Lecture 1          | 59 slides | course overview, derivative definition, market participants, exchange vs OTC, basic option/forward examples |
| Lecture 2 Futures Markets              | 66 slides | futures contract mechanics, margin, clearing house, basis, hedge ratio, stock index futures hedge           |
| Lecture 3 Interest Rates part 1        | 43 slides | rate types, LIBOR/SOFR context, compounding, zero rates, bond pricing, bootstrapping                        |
| Lecture 4 Interest Rates part 2        | 42 slides | forward rates, FRA payoff and valuation, duration and convexity basics                                      |
| Lecture 5 Forward and Futures Prices   | 47 slides | no-arbitrage pricing, investment vs consumption assets, income, dividends, storage cost, convenience yield  |
| Lecture 6 Interest Rate Futures part 1 | 37 slides | day count, Treasury bond quotation, accrued interest, conversion factor, cheapest-to-deliver                |
| Lecture 7 Interest Rate Futures part 2 | 36 slides | Eurodollar futures, duration hedge, convexity adjustment intuition, China rate futures context              |
| Lecture 8 Swaps                        | 48 slides | plain vanilla swaps, comparative advantage, valuation as bonds or FRAs, currency swaps                      |

|                                |           |                                                                                                |
|--------------------------------|-----------|------------------------------------------------------------------------------------------------|
| Lecture 9 Securitization       | 10 slides | ABS, tranches, waterfall, ABS CDO loss allocation                                              |
| Lecture 10 Option Properties   | 49 slides | option payoffs, bounds, early exercise, put-call parity                                        |
| Lecture 11 Trading Strategies  | 37 slides | covered calls, protective puts, spreads, straddles, strips, straps, strangles                  |
| Lecture 12 Binomial Trees      | 34 slides | one-step and two-step binomial pricing, risk-neutral valuation, American option early exercise |
| Supplementary Oil Futures Case | 12 slides | WTI/Brent/SC, term structure, geopolitical shock, backwardation/contango interpretation        |
| Syllabus / 教学实施方案              | 8 pages   | course logistics, textbook, evaluation, reading map                                            |
| Assignments 1-4                | 4 files   | practice question bank and final review problems                                               |
| Assignment 1-3 solutions       | 3 PDFs    | official answer checks for solved practice                                                     |





# 第一章 Big Picture: Derivatives Pricing / 总体框架

Source: Lecture 1; syllabus

## 1.1 什么是衍生品 / What Is a Derivative

**衍生品** (*derivative*) 是一种金融合约，它的价值来自某个 **标的资产** (*underlying asset*)。标的可以是股票、外汇、利率、债券、商品、信用风险，甚至天气指数。课程里的核心工具包括 **forward** (远期)，**futures** (期货)，**swap** (互换)，**option** (期权)，以及基于这些工具组合出来的结构化产品。

### 小白直觉

衍生品不是“凭空赚钱”的工具，而是把未来不确定的价格、利率或现金流变成今天可以交易的合约。pricing 的核心问题是：如果市场没有免费午餐 (no arbitrage)，这个合约今天应该值多少钱？

## 1.2 三类使用者 / Three Types of Users

- **Hedgers** (套期保值者)：目标不是赌方向，而是减少风险。例如航空公司担心油价上涨，可以买入燃油相关 futures。
- **Speculators** (投机者)：主动承担方向风险，希望从价格变化中获利。
- **Arbitrageurs** (套利者)：利用价格关系错位做无风险或近似无风险交易。无套利定价的公式大多来自这个思想。

### 1.3 Exchange vs OTC

|      | Exchange traded                  | OTC                                 |
|------|----------------------------------|-------------------------------------|
| 中文   | 交易所交易                            | 场外交易                                |
| 标准化  | 高，合约规格固定                         | 低，可定制                               |
| 信用风险 | 由 clearing house 降低              | 取决于 counterparty                    |
| 典型例子 | futures, exchange-traded options | forwards, swaps, customized options |

### 1.4 课程主线

这门课的主线可以压缩为四句话：

1. Forward/Futures pricing 用 **cash-and-carry arbitrage** 建立现货和远期价格的关系。
2. Interest rate 部分提供 discounting, zero curve, forward rate, duration 等定价基础。
3. Swaps 可以拆成债券组合或 FRA 组合。
4. Options 不能只用线性复制，必须处理非线性 payoff；put-call parity 和 binomial trees 是入门核心。

## 第二章 Futures Markets and Hedging / 期货市场与套期保值

Source: Lecture 2; Assignment 1

### 2.1 Futures Contract Mechanics

**Futures contract** (期货合约) 是交易所标准化合约。合约会规定:

- underlying asset: 交割什么资产;
- contract size: 一张合约对应多少数量;
- delivery month/place: 何时何地交割;
- price quotation: 价格如何报价;
- daily settlement: 每天按结算价把盈亏打进 margin account。

#### 为什么 futures 要 daily settlement

Forward 的盈亏通常到期一次性结算; futures 每天结算。这样 clearing house 可以把信用风险切成每天的小块。如果价格朝你不利方向变动, margin account 下降到 maintenance margin 以下, 就会触发 margin call。

### 2.2 Margin Account / 保证金

- **Initial margin** (初始保证金): 开仓时先放进去的钱。
- **Maintenance margin** (维持保证金): 账户余额不能低于这个数。
- **Margin call** (追加保证金通知): 余额低于 maintenance margin 时, 要补回 initial margin。

#### Margin call threshold

对于 long futures:

$$\text{margin balance} = \text{initial margin} + (\Delta F) \times \text{contract size} \times N.$$

如果价格下降导致  $\text{balance} < \text{maintenance margin}$ , 就会 margin call。对于 short futures, 价

格上涨导致亏损。

## 2.3 Basis and Convergence

**Basis**（基差）通常定义为

$$\text{basis} = S - F,$$

其中  $S$  是 spot price,  $F$  是 futures price。到交割期附近, futures price 和 spot price 必须收敛, 否则会出现交割套利。Lecture 2 的 convergence 图就是在说明这一点。

## 2.4 Hedging with Futures

最小方差套保比率:

$$h^* = \rho \frac{\sigma_S}{\sigma_F}.$$

这里  $\rho$  是现货价格变化与期货价格变化的相关系数,  $\sigma_S$  是被套保资产价格变化标准差,  $\sigma_F$  是期货价格变化标准差。

要交易的合约数量:

$$N^* = \frac{h^* Q_A}{Q_F},$$

其中  $Q_A$  是需要套保的现货数量,  $Q_F$  是一张 futures contract 的数量。

### 常见错误

很多人会把 hedge ratio 的方向和合约数量混在一起。先判断方向: 担心价格上涨就需要 long futures, 担心价格下跌就需要 short futures; 再用  $N^*$  算张数。

## 2.5 Changing Portfolio Beta

用 stock index futures 改变组合 beta:

$$N^* = (\beta_T - \beta_P) \frac{V_A}{F_0 \times \text{multiplier}}.$$

如果  $N^* > 0$ , 买入 futures; 如果  $N^* < 0$ , 卖出 futures。

## 第三章 Interest Rates / 利率、零息利率与远期利率

Source: Lecture 3; Lecture 4; Assignment 1

### 3.1 Interest Rate Is a Price of Time

利率是时间的价格。今天的 1 元和一年后的 1 元不一样，因为今天的钱可以投资。衍生品定价中最常用的是 **risk-free rate**（无风险利率），它用来 discount 确定性现金流。

### 3.2 Compounding / 复利方式

如果年利率为  $R$ ，每年复利  $m$  次， $T$  年后的金额为

$$A = P \left( 1 + \frac{R}{m} \right)^{mT}.$$

连续复利 (continuous compounding) 为

$$A = Pe^{RT}.$$

离散复利和连续复利转换：

$$R_c = m \ln \left( 1 + \frac{R_m}{m} \right), \quad R_m = m \left( e^{R_c/m} - 1 \right).$$

#### 为什么课程喜欢 continuous compounding

连续复利让很多公式变短。例如 discount factor 是  $e^{-rT}$ ，forward price 是  $S_0 e^{rT}$ 。考试给出 continuously compounded rate 时，通常就是暗示你用指数形式。

### 3.3 Zero Rate and Discount Factor

**Zero rate**（零息利率）是从今天到某个期限的一次性投资收益率。若  $R(T)$  是连续复利 zero rate, discount factor 为

$$P(0, T) = e^{-R(T)T}.$$

有现金流  $C_i$  在  $t_i$  支付，则价格为

$$\text{Price} = \sum_i C_i P(0, t_i).$$

### 3.4 Bootstrapping Zero Curve

Bootstrapping 是用短期限债券价格一步一步推出更长期 zero rates。思路如下：

1. 先用最短期 zero-coupon bond 直接得到第一个 discount factor。
2. 再用有 coupon 的债券价格，扣掉已经知道的早期 coupon 现值。
3. 剩下的部分对应最后一期 principal + coupon，由此解出新的 discount factor。

### 3.5 Forward Rate / 远期利率

如果  $R_1$  是到  $T_1$  的 zero rate,  $R_2$  是到  $T_2$  的 zero rate, 均为连续复利, 则  $T_1$  到  $T_2$  的 forward rate 为

$$R_F = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}.$$

#### Forward rate 的直觉

把钱从 0 存到  $T_2$ ，应该等价于先从 0 存到  $T_1$ ，再从  $T_1$  存到  $T_2$ 。否则就可以借短贷长或借长贷短做套利。

### 3.6 Forward Rate Agreement / FRA

**FRA** (*forward rate agreement*) 是约定未来某段借贷期适用利率的合约。若约定利率为  $R_K$ ，未来市场利率为  $R_M$ ，本金  $L$ ，借贷期长度为  $\tau = T_2 - T_1$ ，则在  $T_2$  的利差现金流近似为

$$L(R_K - R_M)\tau$$

或相反方向，取决于是 lender 还是 borrower。市场惯例通常在  $T_1$  结算，所以要从  $T_2$  折现回  $T_1$ 。

### 3.7 Duration / 久期

**Duration** (久期) 衡量债券价格对收益率变化的敏感性。Modified duration 近似关系：

$$\frac{\Delta B}{B} \approx -D \Delta y.$$

Lecture 6-7 在利率期货套保里使用 duration matching:

$$N^* = \frac{PD_P}{FD_F}.$$

## 第四章 Forward and Futures Pricing / 远期与期货定价

Source: Lecture 5; Assignment 2; supplementary oil futures case

### 4.1 No Arbitrage Pricing

无套利定价的基本句式是：构造两个未来现金流完全一样的组合，它们今天价格必须一样。Forward/Futures pricing 最常见的复制组合是：

- 买入现货并融资持有到期；
- 或签订 forward 到期买入资产。

### 4.2 Investment Asset with No Income

对于不支付收益、无 storage cost 的 investment asset：

$$F_0 = S_0 e^{rT}.$$

如果  $F_0 > S_0 e^{rT}$ ，做 cash-and-carry：借钱买现货，同时 short forward。若  $F_0 < S_0 e^{rT}$ ，做 reverse cash-and-carry：short spot，把钱投资，同时 long forward。

### 4.3 Known Cash Income

如果持有资产期间有确定现金收益，其现值为  $I$ ：

$$F_0 = (S_0 - I) e^{rT}.$$

收益降低 forward price，因为买现货的人在持有期会收到这部分收益。

### 4.4 Known Dividend Yield

如果资产提供连续收益率  $q$ ：

$$F_0 = S_0 e^{(r-q)T}.$$

Stock index futures 常用这个公式，其中  $q$  是 dividend yield。

## 4.5 Foreign Currency

外汇可以看作支付 foreign risk-free rate  $r_f$  的资产：

$$F_0 = S_0 e^{(r_d - r_f)T}.$$

$r_d$  是 domestic interest rate,  $r_f$  是 foreign interest rate。

## 4.6 Commodity Futures: Storage Cost and Convenience Yield

商品分为 **investment assets** (投资资产) 和 **consumption assets** (消费资产)。黄金更像 investment asset, 原油、小麦等更像 consumption asset。若 storage cost 的现值为  $U$ , 无 convenience yield 时：

$$F_0 = (S_0 + U)e^{rT}.$$

若用连续 storage cost rate  $u$  和 convenience yield  $y$ ：

$$F_0 = S_0 e^{(r+u-y)T}.$$

### Convenience yield 的直觉

持有实物商品有时有额外好处，例如炼油厂持有原油可以保证生产不断。这个“持有实物的便利”会让现货更有价值，因此降低 futures price，相当于公式里的  $-y$ 。

## 4.7 Contango and Backwardation

- **Contango** (正向市场)：远期价格高于近月价格，常见原因是 financing/storage cost 较高。
- **Backwardation** (反向市场)：近月价格高于远期价格，常见原因是短期供给紧张、convenience yield 高。

补充油价课件用 WTI、Brent、SC 的 futures curve 说明：地缘冲击会先影响近月合约，因为市场最担心的是即时供给和运输瓶颈，而不是很远未来的均衡价格。



## 第五章 Interest Rate Futures / 利率期货

Source: Lecture 6; Lecture 7; Assignment 2

### 5.1 Day Count Convention

**Day count convention**（日计数惯例）决定利息如何按天累计。

- Treasury bonds: often Actual/Actual。
- Corporate bonds: often 30/360。
- Money market instruments: often Actual/360。

题目里如果给出 quoted price、coupon date、settlement date，必须先判断 accrued interest。

### 5.2 Treasury Bond Quotation

美国 Treasury bond 常用 32 分之一报价。例如 102-07 表示

$$102 + \frac{7}{32} = 102.21875.$$

**Cash price**（现金价格）等于 quoted price 加 accrued interest：

$$\text{Cash price} = \text{Quoted price} + \text{Accrued interest}.$$

### 5.3 Conversion Factor and Cheapest-to-Deliver

Treasury bond futures 允许交割多种债券。为了公平，每只债券有 **conversion factor**（转换因子）。交割时收到：

$$\text{Cash received} = \text{Futures price} \times \text{conversion factor} + \text{accrued interest}.$$

卖方会选择 **cheapest-to-deliver**（最便宜可交割债券），即交割成本最低的债券。

### 5.4 Eurodollar Futures

Eurodollar futures 的报价：

$$Q = 100 - R,$$

其中  $R$  是三个月 LIBOR 的年化百分比。若 quote 为 98.40, 则 implied rate 为 1.60%。

#### Eurodollar hedge direction

借款人担心未来利率上升。利率上升会使 Eurodollar futures price 下降, 所以借款人应该 short Eurodollar futures, 价格下跌时 futures 盈利可以抵消借款成本上升。

## 5.5 Duration Based Hedge

债券组合用 Treasury bond futures 免疫利率风险:

$$N^* = \frac{PD_P}{FD_F}.$$

这里  $P$  是组合价值,  $D_P$  是组合 duration,  $F$  是 futures contract 对应的交割价值,  $D_F$  是 CTD bond duration。若担心利率上升导致债券价格下降, 通常 short futures。

## 第六章 Swaps / 互换

Source: Lecture 8; Assignment 3

### 6.1 Plain Vanilla Interest Rate Swap

**Plain vanilla interest rate swap** (普通利率互换) 是一方支付固定利率、收取浮动利率; 另一方相反。本金通常不交换, 只用 notional principal 计算利息。

#### Swap 的直觉

Swap 可以看成一串 FRA, 也可以看成一只 fixed-rate bond 和一只 floating-rate bond 的组合。哪个角度更方便, 取决于题目给了 swap rate 还是 zero curve。

### 6.2 Using Swaps to Transform Liabilities

如果公司本来借的是 floating rate loan, 又进入 swap 支付 fixed、收 LIBOR, 那么:

$$\text{floating loan payment} + \text{swap net payment}$$

可以合成 fixed rate liability。反过来, fixed loan 加 receive fixed/pay floating swap 可以合成 floating liability。

### 6.3 Comparative Advantage

比较优势题的套路:

1. 分别比较两家公司在 fixed market 和 floating market 的借款差距。
2. 差距更大的市场是有比较优势的市场。
3. 总潜在收益等于两个市场差距之差。
4. 通过 swap 和金融中介分配收益。

## 6.4 Swap Valuation as Bonds

对“pay fixed, receive floating”的一方：

$$V_{\text{swap}} = B_{\text{floating}} - B_{\text{fixed}}.$$

对“receive fixed, pay floating”的一方，符号相反。

## 6.5 Swap Rate and Zero Curve

Par swap rate 是让 swap 初始价值为 0 的 fixed rate。若支付频率为一年一次，notional 为 100，则  $n$  年 par swap 可视为 coupon bond at par：

$$100 = \sum_{i=1}^{n-1} \frac{R_{\text{swap}} \times 100}{(1 + R_i)^i} + \frac{100 + R_{\text{swap}} \times 100}{(1 + R_n)^n}.$$

Assignment 3 第 1 题就是用 2 年和 3 年 swap rate bootstrapping 2 年、3 年 LIBOR zero rates。

# 第七章 Securitization / 证券化

Source: Lecture 9; Assignment 3

## 7.1 Basic Structure

**Securitization** (证券化) 把一组资产的现金流打包给 SPV, 再由 SPV 发行不同层级的证券。  
典型结构:

Loans/Mortgages → SPV → Senior, Mezzanine, Equity tranches.

## 7.2 Waterfall

**Waterfall** (瀑布结构) 决定现金流和损失如何分配。通常收益先给 senior, 再给 mezzanine, 最后给 equity; 损失反过来, equity first loss, 之后 mezzanine, 再之后 senior。

### 为什么 senior tranche 也可能有风险

Senior tranche 不是没有风险, 而是只有在低层级已经被打穿后才开始亏损。ABS CDO 的危险在于它把很多 ABS mezzanine tranches 再打包, 底层相关性一上升, 原本看起来分散的 mezzanine 风险会集中冲击 CDO senior tranche。

## 第八章 Option Properties / 期权性质

Source: Lecture 10; Assignment 1; Assignment 3

### 8.1 Call and Put

- **Call option** (看涨期权): 持有人有权买入标的。
- **Put option** (看跌期权): 持有人有权卖出标的。
- **European option** (欧式期权): 只能到期行权。
- **American option** (美式期权): 到期前任意时间可行权。

### 8.2 Payoff and Profit

Long call payoff:

$$\max(S_T - K, 0).$$

Long put payoff:

$$\max(K - S_T, 0).$$

Profit = payoff - premium accumulated to maturity. 在很多 assignment solution 中, 为了简单忽略 time value of money, 直接用 payoff - premium。

### 8.3 Bounds

不支付股利股票的 European call 下界:

$$c \geq S_0 - Ke^{-rT}.$$

European put 下界:

$$p \geq Ke^{-rT} - S_0.$$

同时  $c \leq S_0$ ,  $p \leq Ke^{-rT}$ 。

## 8.4 Put-Call Parity

无股利 European options:

$$c + Ke^{-rT} = p + S_0.$$

有确定股利现值  $D$ :

$$c + D + Ke^{-rT} = p + S_0.$$

### Parity 的直觉

左边“call + bond + dividend cash reserve”和右边“put + stock”到期现金流一样。到期如果  $S_T > K$ ，两边都等于  $S_T$ ；如果  $S_T < K$ ，两边都等于  $K$ 。今天价格不同就有 arbitrage。

## 8.5 Early Exercise

对于不支付股利的股票，American call 通常不应提前行权。原因是行权要提前支付 strike，损失利息，而且仍然没有获得额外股利。American put 可能提前行权，尤其是 deep in the money 时，因为提前拿到  $K$  可以赚利息。

## 第九章 Trading Strategies / 期权交易策略

Source: Lecture 11; Assignment 3

### 9.1 Covered Call and Protective Put

**Covered call** (备兑看涨): long stock + short call。它牺牲上涨空间, 换取 premium income。

**Protective put** (保护性看跌): long stock + long put。它像给股票买保险, 锁定最低卖出价值。

Put-call parity 可以写成:

$$S_0 + p = c + Ke^{-rT}.$$

这说明 protective put 的经济效果等价于 fiduciary call (call + risk-free bond)。

### 9.2 Spreads

- **Bull spread** (牛市价差): 押温和上涨, 收益和亏损都有限。
- **Bear spread** (熊市价差): 押温和下跌。
- **Butterfly spread** (蝶式价差): 押价格停留在中间 strike 附近。
- **Calendar spread** (日历价差): 同 strike 不同 maturity。

### 9.3 Combinations

- **Straddle** (跨式组合): 同 strike 同 maturity 的 long call + long put, 押大波动。
- **Strip** (条式组合): one call + two puts, 押大波动且更偏向下跌。
- **Strap** (带式组合): two calls + one put, 押大波动且更偏向上涨。
- **Strangle** (宽跨式组合): 不同 strike 的 out-of-the-money call 和 put, 成本更低但需要更大波动。



## 第十章 Binomial Trees / 二叉树定价

Source: Lecture 12; Assignment 4

### 10.1 One-Step Binomial Model

假设股票现在为  $S_0$ ，一个 period 后只可能上涨到  $S_0u$  或下跌到  $S_0d$ 。衍生品 payoff 分别为  $f_u$  和  $f_d$ 。

复制组合 long  $\Delta$  shares and short one derivative，使组合无风险：

$$S_0u\Delta - f_u = S_0d\Delta - f_d.$$

因此

$$\Delta = \frac{f_u - f_d}{S_0u - S_0d}.$$

### 10.2 Risk-Neutral Valuation

更常用的写法是 risk-neutral probability：

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

衍生品价值：

$$f = e^{-r\Delta t} [pf_u + (1-p)f_d].$$

**Risk-neutral probability 不是真实概率**

$p$  不是股票真实上涨概率，而是让 discounted stock price 成为 martingale 的定价权重。考试里不要解释成“市场认为上涨概率是  $p$ ”。

### 10.3 Two-Step Tree

两步树可以从终点向前 backward induction：

$$f_{\text{node}} = e^{-r\Delta t} [pf_{\text{up}} + (1-p)f_{\text{down}}].$$

如果是 American option，每个节点还要比较 continuation value 和 immediate exercise value：

$$f_{\text{American put}} = \max(K - S, e^{-r\Delta t} [pf_u + (1-p)f_d]).$$

## 第十一章 Formula Sheet / 公式速查

| Topic                  | Formula                                 | Use                               |
|------------------------|-----------------------------------------|-----------------------------------|
| Continuous compounding | $A = Pe^{RT}$                           | discounting and future value      |
| Discrete to continuous | $R_c = m \ln(1 + R_m/m)$                | rate conversion                   |
| Forward rate           | $R_F = (R_2T_2 - R_1T_1)/(T_2 - T_1)$   | implied future rate               |
| No income forward      | $F_0 = S_0e^{rT}$                       | investment asset with no income   |
| Known income forward   | $F_0 = (S_0 - I)e^{rT}$                 | stock/bond with known cash income |
| Dividend yield         | $F_0 = S_0e^{(r-q)T}$                   | index futures                     |
| Currency forward       | $F_0 = S_0e^{(r_d - r_f)T}$             | FX forward                        |
| Storage cost           | $F_0 = (S_0 + U)e^{rT}$                 | commodity upper bound             |
| Convenience yield      | $F_0 = S_0e^{(r+u-y)T}$                 | commodity futures                 |
| Minimum variance hedge | $h^* = \rho\sigma_S/\sigma_F$           | cross hedge                       |
| Stock index hedge      | $N^* = (\beta_T - \beta_P)V_A/(F_0M)$   | beta adjustment                   |
| Duration hedge         | $N^* = PD_P/(FD_F)$                     | bond portfolio hedge              |
| Put-call parity        | $c + D + Ke^{-rT} = p + S_0$            | European option arbitrage         |
| Binomial probability   | $p = (e^{r\Delta t} - d)/(u - d)$       | risk-neutral pricing              |
| Binomial value         | $f = e^{-r\Delta t}[pf_u + (1 - p)f_d]$ | option/tree valuation             |

## 第十二章 High-Frequency Practice Questions / 高频重要题

### 12.1 Assignment 1 Q1: Short Forward on GBP

**Question.** An investor enters into a short forward contract to sell 100,000 British pounds at 1.4000 USD/GBP. Gain or loss if final exchange rate is 1.3800 or 1.4300?

**Solution.** Short forward means you must sell GBP at 1.4000. If market is 1.3800, selling at 1.4000 is better:

$$(1.4000 - 1.3800) \times 100000 = 2000.$$

Gain = \$2,000. If market is 1.4300, selling at 1.4000 is worse:

$$(1.4300 - 1.4000) \times 100000 = 3000.$$

Loss = \$3,000.

### 12.2 Assignment 1 Q3: Margin Call

**Question.** A trader buys two July futures contracts on frozen orange juice. Each contract is 15,000 pounds. Current futures price is 200 cents/lb, initial margin \$8,000 per contract, maintenance margin \$6,000 per contract. What price change leads to margin call? When can \$3,000 be withdrawn?

**Solution.** Total initial margin = 16000, total maintenance margin = 12000. Loss that triggers margin call is \$4,000. Two contracts cover 30000 pounds, so price fall:

$$\frac{4000}{30000} = 0.1333 \text{ dollars} = 13.33 \text{ cents.}$$

Margin call when price falls to  $200 - 13.33 = 186.67$  cents/lb. To withdraw \$3,000, account must rise to \$19,000, so gain is \$3,000:

$$\frac{3000}{30000} = 10 \text{ cents.}$$

Price must rise to 210 cents/lb.

## 12.3 Assignment 1 Q6: Cross Hedge

**Question.** A company loses \$1 million for each 1 cent/gallon increase in a new fuel. Correlation with gasoline futures price changes is 0.8. New fuel standard deviation is 50% greater than gasoline futures. Contract size is 40,000 gallons.

**Solution.** Minimum variance hedge ratio:

$$h^* = 0.8 \times 1.5 = 1.2.$$

Exposure measured in gallons:

$$\frac{1000000}{0.01} = 100000000 \text{ gallons.}$$

Because the company loses when fuel price rises, it should gain when gasoline futures rises, so take a long futures position:

$$1.2 \times 100000000 = 120000000 \text{ gallons.}$$

Number of contracts:

$$\frac{120000000}{40000} = 3000.$$

## 12.4 Assignment 1 Q7: Changing Beta

Portfolio value is \$100 million, beta is 1.4, index is 2,000, multiplier is \$250.

**To reduce beta to 0.5:**

$$N = (0.5 - 1.4) \frac{100000000}{2000 \times 250} = -180.$$

Short 180 futures contracts.

**To increase beta to 1.8:**

$$N = (1.8 - 1.4) \frac{100000000}{2000 \times 250} = 80.$$

Long 80 futures contracts.

## 12.5 Assignment 2 Q1: Forward Price with Dividends

Stock price  $S_0 = 100$ , dividends \$1 in 2 months and 5 months,  $r = 8\%$ ,  $T = 0.5$ .

Present value of dividends:

$$I = e^{-0.08(2/12)} + e^{-0.08(5/12)} = 1.9540.$$

Forward price:

$$F_0 = (100 - 1.9540)e^{0.08(0.5)} = 102.05.$$

Initial value of the forward is zero.

Three months later,  $S = 48$ , one dividend remains in 2 months:

$$I = e^{-0.08(2/12)} = 0.9868.$$

The short forward value is

$$f_{\text{short}} = -(S - I - Ke^{-rT}) = 51.05.$$

New 3-month forward price:

$$F = (48 - 0.9868)e^{0.08(3/12)} = 47.96.$$

## 12.6 Assignment 2 Q2: Stock Index Futures

If  $S_0 = 1200$ ,  $r = 9\%$ , average dividend yield  $q = 3.4\%$ , and  $T = 153/365$ :

$$F_0 = 1200e^{(0.09-0.034)(153/365)} \approx 1228.5.$$

## 12.7 Assignment 2 Q3: Currency Forward

Given  $S_0 = 1.4500$ ,  $F_0 = 1.3950$ , USD rate  $r_d = 1\%$ ,  $T = 0.5$ :

$$1.3950 = 1.4500e^{(0.01-r_f)0.5}.$$

So

$$r_f = 0.01 - \frac{\ln(1.3950/1.4500)}{0.5} \approx 8.73\%.$$

## 12.8 Assignment 2 Q4-Q5: Commodity Futures

Oil:  $S_0 = 80$ , storage cost \$3 paid at the end of one year,  $r = 5\%$ :

$$F_0 \leq 80e^{0.05} + 3 = 87.10.$$

It is an upper bound because oil is a consumption asset and convenience yield may reduce futures price.

Silver:  $S_0 = 15$ , storage \$0.24 per year paid quarterly in advance,  $T = 0.75$ ,  $r = 10\%$ :

$$U = 0.06 + 0.06e^{-0.10(0.25)} + 0.06e^{-0.10(0.5)} = 0.1756.$$

$$F_0 = (15 + 0.1756)e^{0.10(0.75)} \approx 16.36.$$

## 12.9 Assignment 2 Q8: Eurodollar Futures Hedge

December Eurodollar futures quote is 98.40, so implied LIBOR is  $100 - 98.40 = 1.60\%$ . Borrowing at LIBOR + 0.5% locks in about 2.10%.

A borrower should short Eurodollar futures. If actual 3-month rate is 1.3%, final settlement price is

$$100 - 1.3 = 98.70.$$

## 12.10 Assignment 3 Q1: Swap Rates to Zero Rates

One-year LIBOR zero rate is 10% annually compounded. Two-year swap rate is 11%. Treat the par swap as a par coupon bond:

$$\frac{11}{1.10} + \frac{111}{(1 + R_2)^2} = 100.$$

Thus  $R_2 = 11.05\%$ .

For the 3-year swap rate 12%:

$$\frac{12}{1.10} + \frac{12}{(1.1105)^2} + \frac{112}{(1 + R_3)^3} = 100.$$

Thus  $R_3 = 12.17\%$ .

## 12.11 Assignment 3 Q5: ABS CDO Loss Waterfall

Mortgage portfolio loss is 16%. ABS has equity 5%, mezzanine 20%, senior 75%.

ABS losses:

- Equity tranche: 100%.
- Mezzanine tranche:  $(16 - 5)/20 = 55\%$ .
- Senior tranche: 0%.

ABS CDO is created from mezzanine tranches, so its collateral loss is 55%. With CDO equity 5%, mezzanine 20%, senior 75%:

- CDO equity: 100%.
- CDO mezzanine: 100%.
- CDO senior:  $(55 - 25)/75 = 40\%$ .

## 12.12 Assignment 3 Q6-Q7: Option Profit Conditions

Long European put: premium \$3, strike \$40. Exercise if  $S_T < 40$ . Profit if

$$40 - S_T - 3 > 0 \Rightarrow S_T < 37.$$

Short European call: premium \$4, strike \$50. The option holder exercises if  $S_T > 50$ . Seller profit is

$$4 - \max(S_T - 50, 0).$$

Seller profits if  $S_T < 54$ .

### 12.13 Assignment 3 Q8: Put-Call Parity Arbitrage

Call and put both cost \$3,  $K = 20$ ,  $T = 0.25$ ,  $r = 10\%$ ,  $S_0 = 19$ , dividend \$1 in one month.

Parity says

$$c + D + Ke^{-rT} = p + S_0.$$

Left side:

$$3 + e^{-0.10/12} + 20e^{-0.10(0.25)} \approx 23.50.$$

Right side:

$$3 + 19 = 22.$$

Left side is overpriced. Arbitrage: sell call, borrow present value of strike, buy put and stock. The future stock/dividend cash flows repay the synthetic obligations and lock in the mispricing.

### 12.14 Assignment 3 Q9-Q10: Butterfly Spread

Put prices for strikes 55, 60, 65 are 4, 6, 9. Buy one 55 put, buy one 65 put, sell two 60 puts.

Initial cost:

$$4 + 9 - 2(6) = 1.$$

Net profit is:

$$\begin{cases} -1, & S_T \leq 55, \\ S_T - 56, & 55 < S_T \leq 60, \\ 64 - S_T, & 60 < S_T \leq 65, \\ -1, & S_T > 65. \end{cases}$$

Loss occurs when  $S_T < 56$  or  $S_T > 64$ . Put-call parity shows the cost is identical to the same butterfly built with calls because the bond and stock terms cancel when combining strikes symmetrically.

### 12.15 Assignment 4 Q1-Q2: Two-Step Binomial Tree

This assignment has no official solution PDF in the materials. The following is a non-official solution derived from Lecture 12.

Given  $S_0 = 50$ ,  $u = 1.06$ ,  $d = 0.95$ ,  $\Delta t = 0.25$ ,  $r = 5\%$ ,  $K = 51$ :

$$p = \frac{e^{0.05(0.25)} - 0.95}{1.06 - 0.95} = 0.568895.$$

Terminal stock prices:

$$S_{uu} = 56.18, \quad S_{ud} = 50.35, \quad S_{dd} = 45.125.$$

Call payoffs:

$$5.18, \quad 0, \quad 0.$$

European call value:

$$c = e^{-0.05(0.5)} [p^2(5.18)] \approx 1.64.$$

Put payoffs:

$$0, \quad 0.65, \quad 5.875.$$

European put:

$$p_{\text{put}} \approx 1.38.$$

Put-call parity check:

$$c + Ke^{-0.05(0.5)} = 1.635 + 49.741 \approx 51.376,$$

$$p + S_0 = 1.376 + 50 = 51.376.$$

For American put, compare continuation and early exercise. At the down node, immediate exercise is  $51 - 47.5 = 3.5$ , continuation value is about 2.87, so early exercise is optimal there. American put value is about 1.65.

## 12.16 Assignment 4 Q3: One-Step Derivative Paying $S_T^2$

This problem uses  $S_0 = 25$ ,  $S_u = 27$ ,  $S_d = 23$ ,  $r = 10\%$ ,  $T = 2/12$ . Here  $u = 27/25$ ,  $d = 23/25$ :

$$p = \frac{e^{0.10(2/12)} - 23/25}{27/25 - 23/25} = 0.60504.$$

Payoff is  $S_T^2$ , so

$$f_u = 27^2 = 729, \quad f_d = 23^2 = 529.$$

Value:

$$f = e^{-0.10(2/12)} [0.60504(729) + (1 - 0.60504)(529)] \approx 639.26.$$



## Final Exam Checklist / 考前检查

- Can you explain no-arbitrage without memorizing?
- Can you convert continuous and semiannual rates?
- Can you bootstrap a zero rate from bond/swap prices?
- Can you price a forward with income, dividend yield, storage cost, or FX yield?
- Can you choose long vs short futures for a hedge?
- Can you calculate margin call price levels?
- Can you compute stock index beta hedge contracts?
- Can you compute Eurodollar implied rate from quote?
- Can you value a swap as fixed bond vs floating bond?
- Can you allocate losses through ABS/CDO tranches?
- Can you write put-call parity with dividends?
- Can you draw and interpret option strategy payoffs?
- Can you run a binomial tree backward?
- Can you decide American put early exercise?